Your Own AWS Account

If you're completing this workshop at your own pace, you can use your personal AWS account. This guide will help you set up the necessary permissions and configure AWS CLI for use with Kiro IDE.

> **AWS Costs**
> You are responsible for the cost of AWS services used while running this workshop in your AWS account. Most resources used in this workshop are eligible for the AWS Free Tier, but charges may apply depending on your account age and usage patterns.

Prerequisites

Before proceeding, ensure you have:

1. An active AWS account with console access
2. AWS CLI installed on your local machine
3. Kiro IDE installed and authenticated

Step 1: Create Workshop IAM Role

We'll use a CloudFormation template to automatically create an IAM role with the necessary permissions for this workshop.

> **The CloudFormation template creates**
> - An IAM role named `kiro-workshop-participant-role`
> - PowerUserAccess permissions (full access to AWS services except IAM and Organizations management)
> - Trust policy allowing you to assume the role from your AWS account

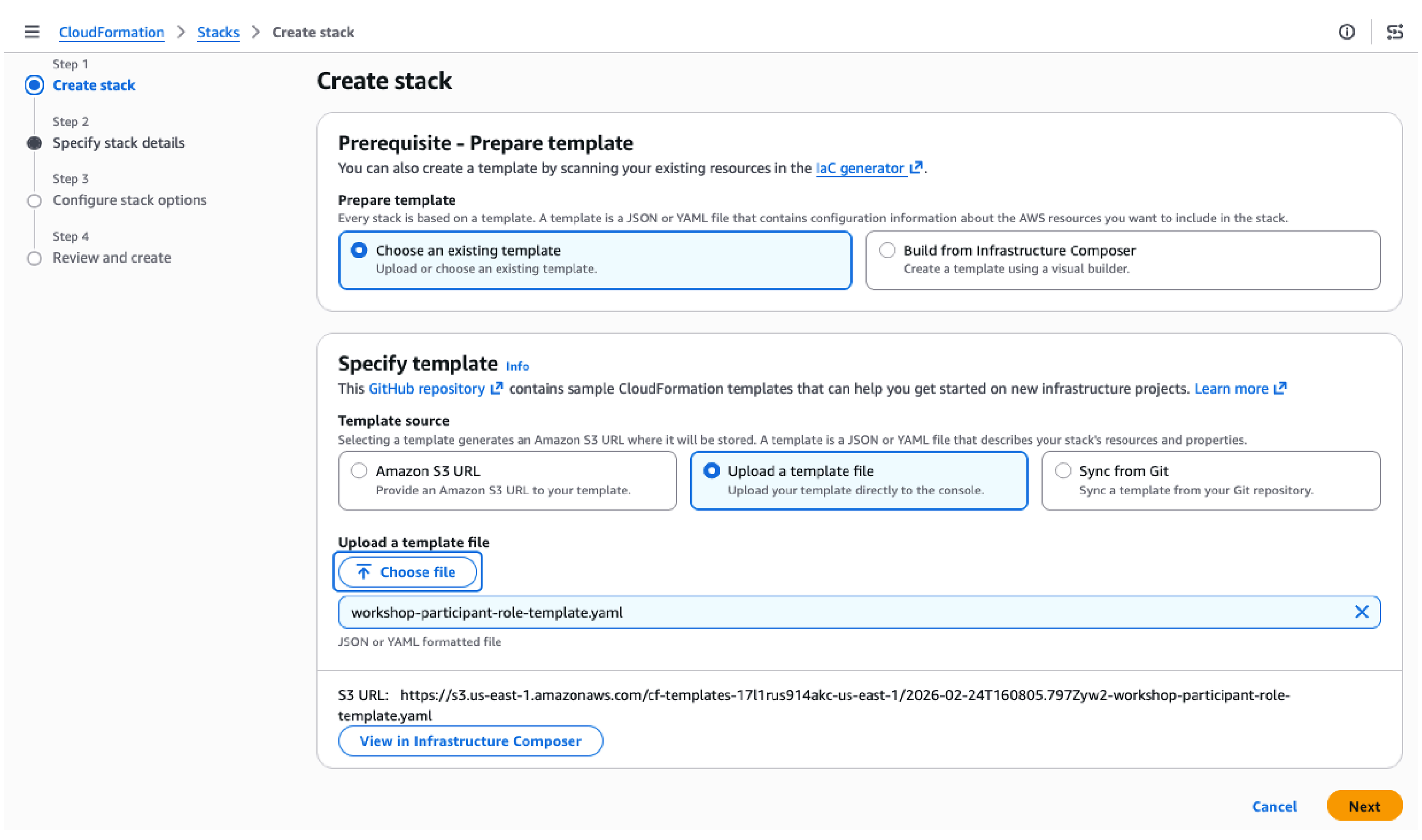
Download the CloudFormation Template

First, download the CloudFormation template to your local machine:

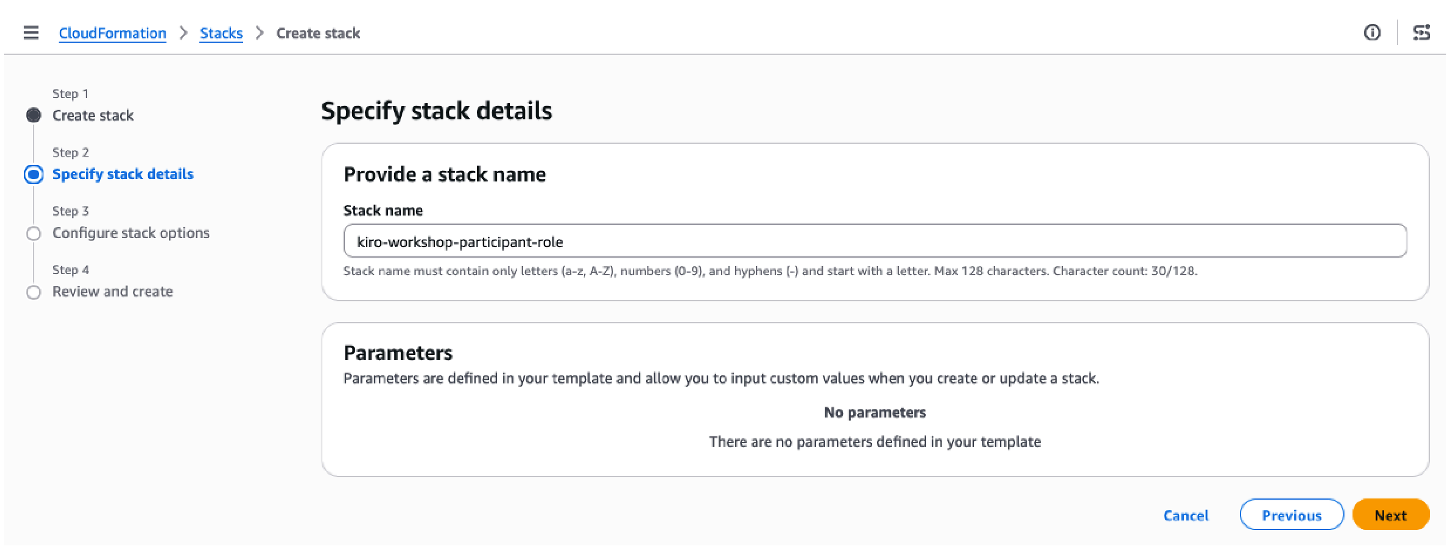
Download Template

Deploy the Stack

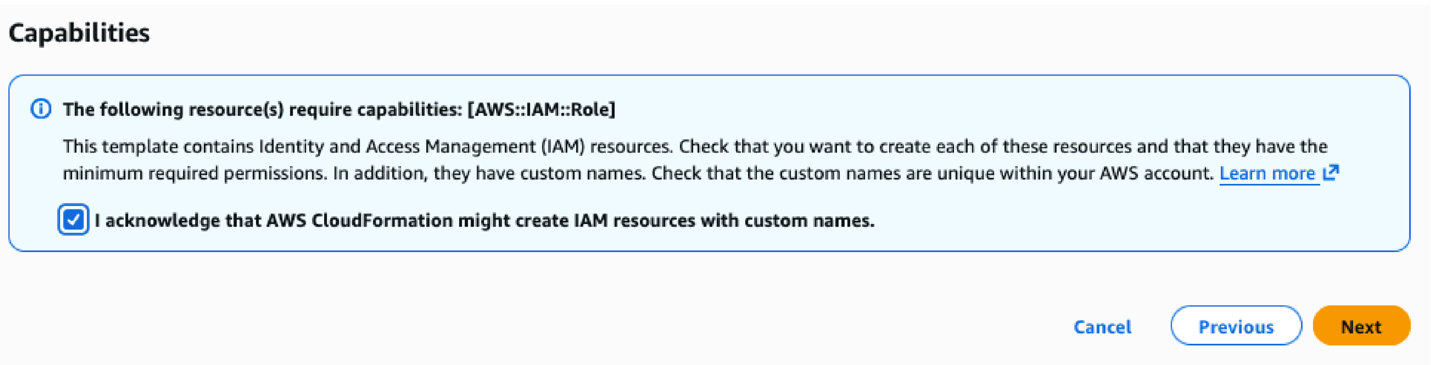
1. Open the CloudFormation Console in us-east-1.
2. Choose **Upload a template file**.
3. Click **Choose file** and select the downloaded `workshop-participant-role-template.yaml`.
4. Click **Next**.



5. For **Stack name**, enter: `kiro-workshop-participant-role`



6. Click **Next** twice to reach the Review page.
7. Scroll to the bottom and check the box: **I acknowledge that AWS CloudFormation might create IAM resources**.

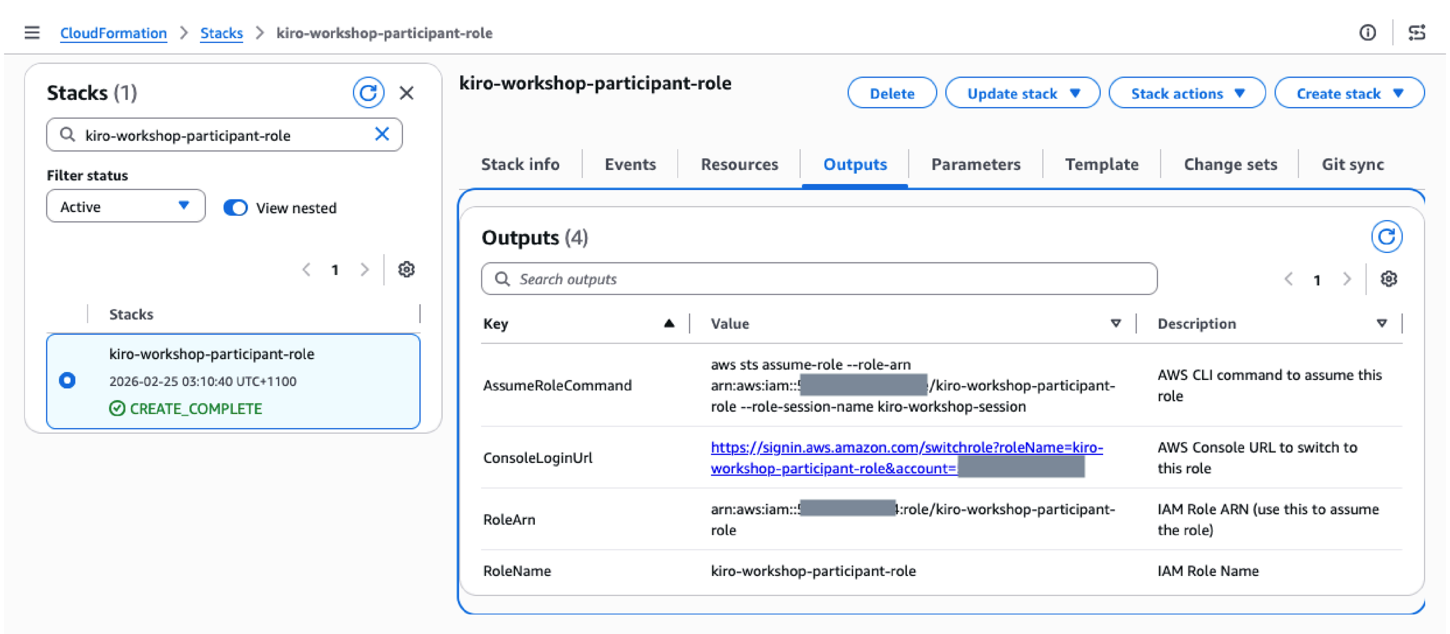


8. Click **Submit**.
9. Wait for stack creation to complete (1-2 minutes). The status will change from `CREATE_IN_PROGRESS` to `CREATE_COMPLETE`.

Retrieve Role Information

Once the stack is created, retrieve your role details:

1. In the CloudFormation console, filter the `kiro-workshop-participant-role` stack name.
2. Click the **Outputs** tab.



3. Copy the **RoleArn** value - you'll need it in the next step. It will look like:

```
arn:aws:iam::123456789012:role/kiro-workshop-participant-role
```

> **What You'll See**
> The Outputs tab shows:
> - **RoleName**: kiro-workshop-participant-role
> - **RoleArn**: The full ARN you'll use for AWS CLI configuration
> - **AssumeRoleCommand**: Ready-to-use CLI command for manual role assumption
> - **ConsoleLoginUrl**: URL to switch to this role in the AWS Console

Step 2: Configure AWS CLI to Use the Role

Now configure AWS CLI to assume the workshop role. You'll need your existing AWS credentials (from your root or admin user) to assume the role.

1. Open your AWS config file in a text editor:
   - **Windows**: `C:\Users\<USERNAME>\.aws\config`
   - **Linux/Mac**: `~/.aws/config`
2. Add a new profile for the workshop role (replace `YOUR_ACCOUNT_ID` with your actual 12-digit account ID from the RoleArn):

```
[profile kiro-workshop]
role_arn = arn:aws:iam::YOUR_ACCOUNT_ID:role/kiro-workshop-participant-role
source_profile = default
region = us-east-1
output = json
```

3. Save the file.

▶ **Alternative: Use Environment Variables**

Step 3: Verify Configuration

1. Test your AWS CLI configuration in Kiro IDE by running:

```
aws sts get-caller-identity --profile kiro-workshop
```

2. Verify the output shows you've assumed the role. You should see:

```
{
    "UserId": "AIDACKCEVSQ6C2EXAMPLE:kiro-workshop-session",
    "Account": "123456789012",
    "Arn": "arn:aws:sts::123456789012:assumed-role/kiro-workshop-participant-role/kiro-workshop-session"
}
```

> **Configuration Complete**
> Your Kiro IDE is now configured to use the workshop role. You can proceed with the workshop exercises.
>
> **Note**: When running AWS CLI commands in the workshop, use `--profile kiro-workshop` or set `AWS_PROFILE=kiro-workshop` as an environment variable.

Best Practices

Security Recommendations

1. **Use IAM Users, Not Root**: Never use your AWS root account credentials for daily operations
2. **Enable MFA**: Add multi-factor authentication to your IAM user for enhanced security
3. **Rotate Access Keys**: Regularly rotate your access keys (every 90 days recommended)
4. **Use Least Privilege**: Grant only the permissions needed for the workshop
5. **Monitor Usage**: Regularly check AWS Cost Explorer and CloudWatch for unexpected activity

Cost Management

1. **Set Up Billing Alerts**: Configure CloudWatch billing alarms to notify you of unexpected charges
2. **Use AWS Budgets**: Create a budget to track workshop spending
3. **Clean Up Resources**: Delete all workshop resources when finished to avoid ongoing charges
4. **Review Free Tier**: Understand which services and usage levels are covered by the Free Tier

▶ **Setting Up Billing Alerts**

Troubleshooting

▶ **AWS CLI Not Found**

▶ **Invalid Credentials Error**

▶ **Permission Denied Errors**

Next Steps

With your personal AWS account configured:

1. Verify you can access AWS services through Kiro
2. Review the workshop prerequisites
3. Proceed to the workshop exercises

> **Remember**
> You are responsible for any AWS costs incurred. Make sure to clean up all resources at the end of the workshop to avoid ongoing charges.